A DERIVED COMPLETE MODULE ANNIHILATED BY A FINITELY GENERATED IDEAL IS ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

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Editorial note. The counted proof span contains the author's somewhat terse argument 'If i does not exist, then M = 0 which is what we want to show. Then N is derived complete ...'; that wording, delivered verbatim, includes the author's ordering of the two cases. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The derived complete modules form a Serre subcategory, hence are closed under cokernels. invoked at proof lines 26110-26112. Shows the partial quotient N is again derived complete.
- Derived completeness of a module M with respect to a finitely generated ideal, formulated as the vanishing of R lim of the torsion system. used at proof lines 26114-26118. Turns the nonzero limit of the constant system into a contradiction.
- The finite sequence of partial quotients M/(f_1, ..., f_i)M. used at proof lines 26107-26109. Provides the index i at which the reduction to one element is made.

Source reference. Original lemma 26068: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26068–26068. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let I be a finitely generated ideal of a ring A. Let M be a derived complete A-module. If $M/IM = 0$, then $M = 0$.*

Proof. Assume that M/IM is zero. Let $I = (f_1, \ldots, f_r)$. Let $i < r$ be the largest integer such that $N = M/(f_1, \ldots, f_i)M$ is nonzero. If i does not exist, then $M = 0$ which is what we want to show. Then N is derived complete as a cokernel of a map

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between derived complete modules, see Lemma 26068. By our choice of i we have that $f_{i+1} : N \to N$ is surjective. Hence

$$\lim(\ldots \to N \xrightarrow{f_{i+1}} N \xrightarrow{f_{i+1}} N)$$

is nonzero, contradicting the derived completeness of N. $\square$